Group 9

Event Detection and Video Summarization

Pipeline Overview Presentation

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Overview of Videos

Video 33 — A speaker shares a story to inspire an audience, followed by a step team performance.

Video 34 — A large group breaks into a choreographed dance, drawing an energetic crowd.

Video 35 — Musicians perform classical music, surprising and delighting commuters.

Video 36 — A beekeeper handles hives and demonstrates how honey is extracted.

Pipeline Overview: Six Step Process

STEP 1: ANNOTATE

Mark salient events with timestamps and importance ratings (-2 to +2).

Provides ground truth for all downstream evaluation.

STEP 2: INTER ANNOTATOR AGREEMENT

Compute timeline κ, saliency κ, and mean IoU across annotators.

Quantifies expected human agreement before modelling.

STEP 3: VLM EVENT DETECTION

SmolVLM2-2.2B samples 6 frames per video.

Outputs event descriptions via 3 prompting strategies.

Annotations and Inter Annotator Agreement

METRICS OVERVIEW

Timeline κ: 0.15 – 0.48
Saliency κ: -0.235 – 0.60 (mean 0.23)
Boundary mIoU: 0.55 – 0.72

→ Boundaries agree; counts & saliency don't.

SUBJECTIVITY ISSUES

Video 36 (beekeeper): κ = -0.235 (worse than random chance)

Narrative vs. visual priorities irreconcilable — no central climax.

BOUNDARY AGREEMENT

mIoU 0.55 – 0.72 across all 4 videos.

When annotators notice an event, they draw similar windows.

VLM Event Detection: SmolVLM2 Performance

MODEL SELECTION

SmolVLM2-2.2B over Qwen2.5.

Native video/frame support;
fits within 8 GB VRAM budget.

PROMPTING STRATEGIES

direct/events → 4–5 events/video, cov@0.5 ≤ 30% ✓

caption-aggregate → video 36: cov@0.5 = 42% ✓

direct/timed → 0% on 3 of 4 videos ✗

FAILURE MODES

① Dialogue events lost (no ASR)

② Model loops / repeats
Fix: repetition_penalty = 1.3
no_repeat_ngram = 4

Moment Retrieval: CLIP Backbone Comparison

CLIP PERFORMANCE

CLIP-L/14: mIoU = 0.113
CLIP-B/32: mIoU = 0.089

L/14 outperforms by +27%

RECALL METRICS

R@0.3: L/14 = 0.211 | B/32 = 0.140
R@0.5: L/14 = 0.088 | B/32 = 0.070

Paraphrase fusion: no gain

DIALOGUE LIMITATIONS

Performance: mIoU = 0.174
Beekeeping: mIoU = 0.070
Speech / dialogue: mIoU = 0.000

CLIP is text-blind — ASR needed

Video 36 Walkthrough

19 GT events (15 from annotator A, 11 from annotator B, merged) across ~4 min of beekeeper interview. Best VLM strategy: caption-aggregate — 1 broad prediction, cov@0.5 = 42.1%, mean sim 0.491. Works because a single generic description captures the 'beekeeper + honey' gist and 8 of 19 events align with it semantically. CLIP-L/14 beekeeping mean IoU = 0.070: opening hives, showing frames, and honey extraction look visually identical to CLIP. Generated summary: 40 s. Saliency κ = -0.235 — annotators disagreed fundamentally on what mattered, which flows directly into the findings on slide 8.

Findings and Critical Reflections

SUBJECTIVITY IN SUMMARIZATION

Subjectivity shapes the summary. On video 36 (beekeeper, saliency $\kappa = -0.235$), annotators had incompatible mental models of salience. CLIP then picks whichever events are most visually distinctive — not most narratively important — so the generated summary reflects neither annotator's intent coherently.

IMPACT OF SALIENCY

Per-category mean IoU (CLIP-L/14): performance/dancing = 0.174, beekeeping = 0.070, speech/dialogue = 0.000 (9 events, zero recall on both backbones). Neutral-saliency events dip to 0.027 — transitional moments with no distinctive visual anchor are invisible to CLIP.

EVENT DETECTION PERFORMANCE

VLM event detection queries yielded improved matching for distinctive events, achieving a mean IoU of **0.206**, highlighting the potential of these methods despite challenges with dialogue events.

Future Work: Planned Improvements

ASR ENHANCEMENT

Whisper-tiny ASR is the highest-priority next step. Dialogue/speech events score 0.000 mean IoU on all methods — 9 events are completely unrecoverable without transcripts. Adding ASR is the single change expected to produce the largest absolute gain in coverage and retrieval.

SALIENCY REINTRODUCTION

Reintroducing saliency through LLM judge re-ranking will enhance the **coherence of generated summaries**, ensuring that the most relevant content is prioritized for better narrative quality.

CG DETR RESTORATION

Once a working checkpoint becomes available, we will restore the **CG DETR comparison**, allowing us to evaluate and improve our moment retrieval capabilities effectively.